GENERATIVE AI FOR EARTH OBSERVATION

Generative AI: Fundamentals, LLMs and NLP for Geodata Processing

Foundations of generative models and large language models, and how they are reshaping the way we query, interpret, and interact with geospatial data.

An Introduction for Students & Researchers in Geoinformatics

ROADMAP

What We'll Cover

01

Foundations

Generative AI, deep learning & NLP building blocks

02

Transformers & LLMs

Architecture, training, and how language models learn

03

Prompting & RAG

In-context learning and retrieval-augmented generation

04

Geodata Applications

Text-to-spatial-SQL, chatbots, remote sensing & metadata

05

Challenges

Hallucination, spatial reasoning limits & open problems

06

The Road Ahead

Geospatial foundation models & future directions

PART 1

Foundations of Generative AI

From artificial intelligence to models that create

CORE CONCEPT

What Is Generative AI?

Generative AI refers to models that learn the underlying patterns and structure of training data well enough to **produce new, original content** — text, images, code, or even spatial data — rather than simply classifying or labeling existing data.

Discriminative Models

Learn a decision boundary to classify or predict labels from input data.

e.g. “Is this pixel water or land?”

Generative Models

Learn the full data distribution and can sample new, realistic outputs from it.

e.g. “Describe this satellite scene in natural language.”

Generative AI Spans Many Modalities

Text — LLMs (GPT, Llama, Qwen) — chat, summarization, code, Q&A

Images — Diffusion models — synthetic imagery, super-resolution

Geodata — Text-to-SQL, scene captioning, spatial reasoning over maps

Audio / Video — Speech synthesis, video generation, simulation

CONTEXT

From AI to Generative AI: A Quick Timeline

Artificial Intelligence

Broad field: machines performing tasks that need human-like intelligence

→

Machine Learning

Systems that learn patterns from data instead of hardcoded rules

→

Deep Learning

Multi-layer neural networks learning hierarchical representations

→

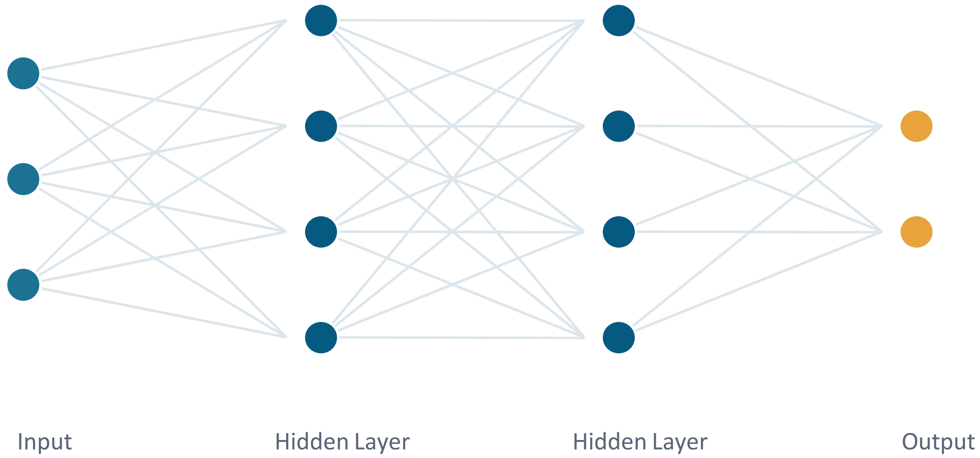
Generative AI / LLMs

Models that generate new content, incl. large language models

Each layer is a subset of the one before it: Generative AI (and LLMs specifically) sit inside deep learning, which sits inside machine learning, which sits inside the broader field of AI.

Neural Networks: The Engine Underneath

- Neural networks are layers of simple math units (“neurons”) that combine weighted inputs and pass them through a non-linear activation function.
- Training adjusts millions (or billions) of weights so the network's output matches expected results, using backpropagation and gradient descent.
- Deep learning stacks many such layers, letting the network learn increasingly abstract features — from edges/tokens to full concepts.
- For geodata: a network can learn to map raw pixels or coordinates directly to labels, embeddings, or generated descriptions.



NLP Fundamentals: Turning Text into Numbers

01

Tokenization

Text is split into sub-word units (tokens) — e.g. “geoprocessing” → [“geo”, “process”, “ing”]

→

02

Embeddings

Each token is mapped to a dense numeric vector capturing semantic meaning — similar words land close together

→

03

Language Modeling

The model learns to predict the next token given prior context, across billions of training sequences

→

04

Contextual Representation

Modern models (transformers) let each token's vector depend on the entire surrounding context, not just itself

PART 2

Transformers & Large Language Models

The architecture that made modern LLMs possible

The Transformer: Attention Is the Key Idea

- Introduced in 2017 (“Attention Is All You Need”), the transformer replaced recurrence with a self-attention mechanism.
- Self-attention lets every token weigh and incorporate information from every other token in the sequence — in parallel.
- This enables efficient training on massive corpora and captures long-range dependencies (e.g. linking a query term to a schema field far away in a prompt).
- Stacking many attention + feed-forward blocks builds the deep networks behind GPT, Llama, Qwen, and similar LLMs.

DEFINITION

What Makes a Language Model "Large"?

Billions

of parameters — the learned weights that encode linguistic and world knowledge

Trillions

of tokens seen during pretraining, drawn from text, code, and other corpora

Emergent

abilities like reasoning and few-shot learning appear only at sufficient scale

Small (task-specific) models vs. Large Language Models

	SMALL MODEL	LARGE LANGUAGE MODEL
Scope	Narrow, single-task (e.g. classify a land-cover pixel)	Broad, general-purpose across many tasks and domains
Training data	Small, curated, task-labeled datasets	Massive, diverse web-scale + domain corpora
Adaptation	Retrain from scratch per task	Prompted, fine-tuned, or retrieval-augmented for new tasks

TRAINING PIPELINE

How an LLM Is Built, Stage by Stage

Pretraining

Self-supervised next-token prediction over massive, diverse text/code corpora — builds general language & world knowledge.

→

Fine-Tuning

Supervised training on curated instruction/response pairs to make the model follow instructions reliably.

→

Alignment (RLHF)

Human feedback trains a reward model that further tunes outputs toward helpful, accurate, safe responses.

→

Domain Adaptation

Optional further tuning or retrieval grounding on domain data — e.g. geospatial schemas & terminology.

Prompting & In-Context Learning

- Pretrained LLMs can perform new tasks purely from a well-crafted prompt — no weight updates required.
- Zero-shot: task described only in instructions. Few-shot: a handful of examples are shown in the prompt itself.
- Chain-of-thought prompting asks the model to reason step-by-step, improving accuracy on multi-step tasks (e.g. building a spatial query).
- System prompts can encode domain context — schema definitions, coordinate systems, naming conventions — to steer outputs toward valid geodata operations.

EXAMPLE: FEW-SHOT PROMPT

```
System: You convert natural-language questions into
PostGIS SQL using the given schema.

Example: “Rivers within 5 km of Dehradun” →
SELECT * FROM rivers WHERE ST_DWithin(geom, :dehradun,
5000);

Query: “Forest patches larger than 10 sq. km in
Uttarakhand” →
```

```
SELECT * FROM forest_patches
WHERE state='Uttarakhand' AND area_sqkm > 10;
```

GROUNDING TECHNIQUE

Retrieval-Augmented Generation (RAG)

LLMs have fixed, frozen knowledge from training and can hallucinate facts. RAG grounds responses in real, up-to-date, or domain-specific data — retrieved at query time and inserted into the prompt.

User Query

“What's the elevation profile near the study site?”

→

Retrieve

Search a geospatial vector/metadata store for relevant records or documents

→

Augment

Inject retrieved context (coordinates, layers, metadata) into the LLM prompt

→

Generate

LLM produces a grounded, source-backed natural-language or SQL answer

PART 3

LLMs & NLP in Geodata Processing

Where language models meet maps, layers, and spatial queries

THE CORE CHALLENGE

Why Geodata Is Hard for General-Purpose LLMs

Spatial Reasoning

Concepts like “near,” “within,” “adjacent to” require geometric reasoning most LLMs never explicitly learned.

Complex Schemas

Geodatabases (PostGIS, shapefiles, rasters) have domain-specific schemas, spatial indexes, and coordinate reference systems.

Multiple Data Types

Vector, raster, point-cloud, and imagery data each need different handling — plain text models see none of it natively.

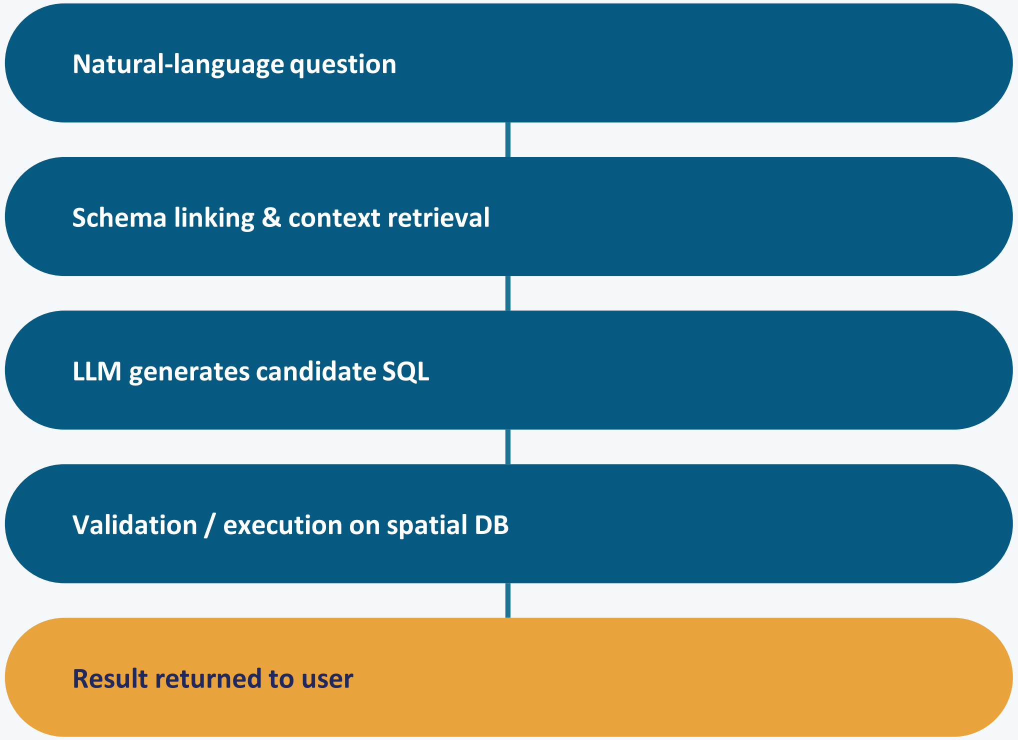
Precision Matters

A subtly wrong query (wrong SRID, wrong buffer distance) can silently return plausible-looking but incorrect results.

Natural Language to Spatial Query (Text-to-SQL)

- Text-to-SQL systems translate a natural-language question directly into an executable spatial query (e.g. PostGIS SQL).
- Schema linking maps question terms (“river,” “elevation”) to the correct tables, columns, and spatial functions.
- Smaller, fine-tuned or well-prompted LLMs (sub-10B parameters) can achieve strong accuracy on this task when grounded with schema context — making on-premise, cost-efficient deployment feasible.
- Reduces the barrier for non-technical users (researchers, decision-makers) to query large geospatial databases without writing SQL.

PIPELINE



LLM & NLP Use Cases Across the Geodata Workflow

Conversational GIS Assistants

Chat interfaces that let users explore maps, layers, and datasets in plain language.

Remote-Sensing Captioning

Generating natural-language descriptions or summaries of satellite/aerial scenes.

Automated Metadata Generation

Drafting FAIR-compliant metadata, abstracts, and documentation for datasets.

Semantic Search over Archives

Embedding-based retrieval across large geospatial data catalogs and reports.

Geocoding & Entity Resolution

Extracting and disambiguating place names and coordinates from unstructured text.

Disaster & Risk Reporting

Summarizing sensor/field reports into structured situational briefs.

A Glimpse from Applied Research

Conversational Geospatial Query Systems

Research prototypes such as chatbot-style interfaces for geospatial data (e.g. GeoChatbot-style systems) explore how sub-10B-parameter LLMs can be paired with schema-aware prompting to translate natural-language questions into geospatial SQL — an approach also explored in recent applied research on text-to-SQL for GIS databases.

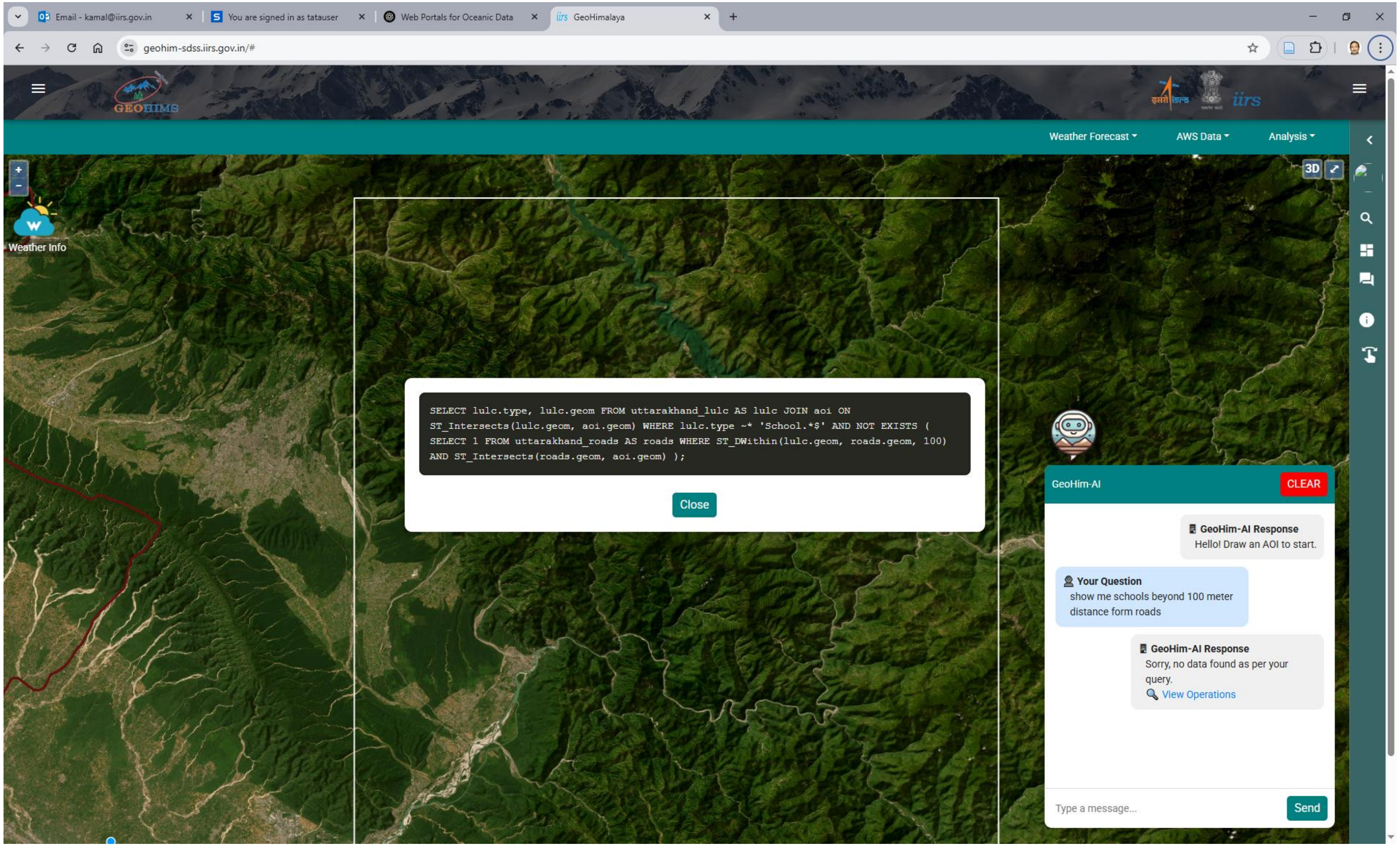
- Smaller, efficient models can be viable when grounded with the right schema and prompt context.
- Hallucination and schema-linking accuracy remain the key evaluation challenges.
- On-premise deployment is attractive for sensitive or infrastructure-hosted government geodata.

Prompting & In-Context Learning – Example Use Case

Automation of Extracting Information from Large Geodatabase using Large Language Models

Information to be extracted	Context GIS Layer	GIS Query
Locate all the high schools	• Schools	SELECT * FROM schools WHERE type = 'High School';
Find all the water bodies within 100 meters distance from roads	• Road • Water Bodies	SELECT DISTINCT w.* FROM water_bodies AS w JOIN roads AS r ON ST_DWithin(w.geom, r.geom, 100)
Show the schools which are within 100 meters distance from road and waterbody	• Road • Water body • Schools	SELECT DISTINCT s.* FROM schools AS s JOIN roads AS r ON ST_DWithin(s.geom, r.geom, 100) -- 100 meters to any road JOIN water_bodies AS w ON ST_DWithin(s.geom, w.geom, 100) -- 100 meters to any water body

Prompting & In-Context Learning – Example Use Case



Natural Language GIS Queries using LLM –QGIS Plugin

- Mistral:7b-instruct is an instruction-tuned open-source LLM optimized for task-based responses.

- **Features**:
  - 7B parameters → lightweight compared to GPT-4 (175B)
  - Open-source & fast: runs locally via Ollama
  - Trained to follow instructions (ideal for prompt-based GIS queries)
  - Suitable for QGIS plugin integration.
  - LightweightLocal (no cloud dependency)
  - Fast response time

Natural Language GIS Queries using LLM –QGIS Plugin

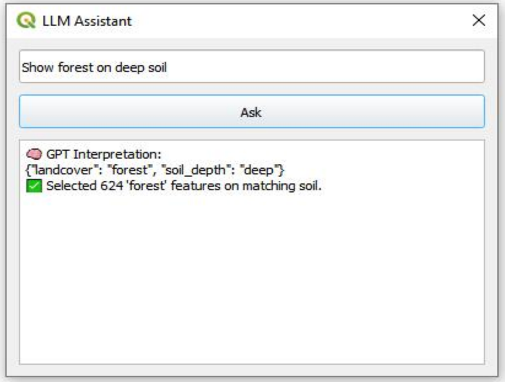
Examples

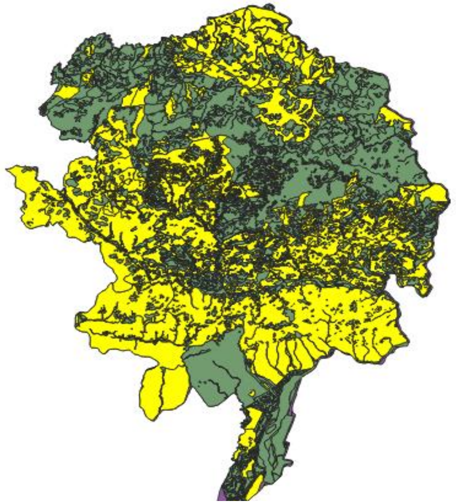
Natural Language Query:
Show forest on deep soil

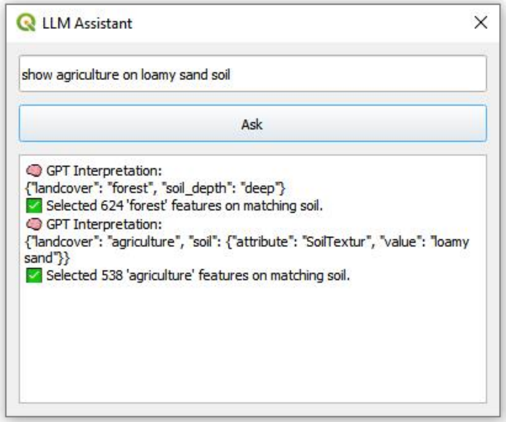
LLM Response:
jsonCopyEdit{ "landcover": "forest", "soil_depth": "deep"}

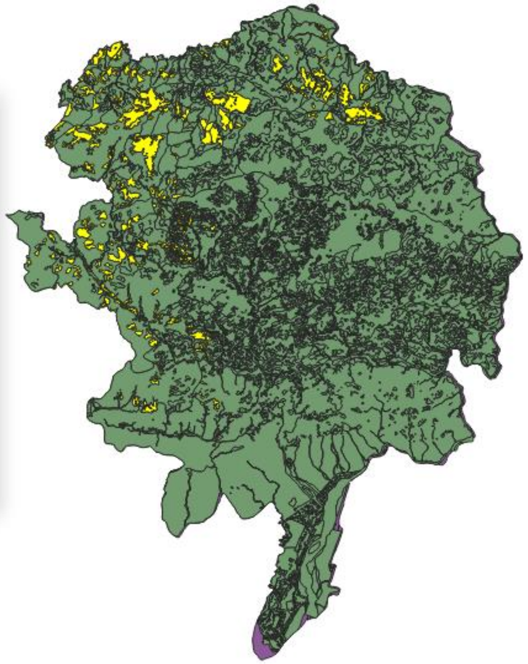
Natural Language Query:
Show agriculture on loamy sand soil

LLM Response:
jsonCopyEdit{ "landcover": “agriculture", "soil_textur": “sand"}





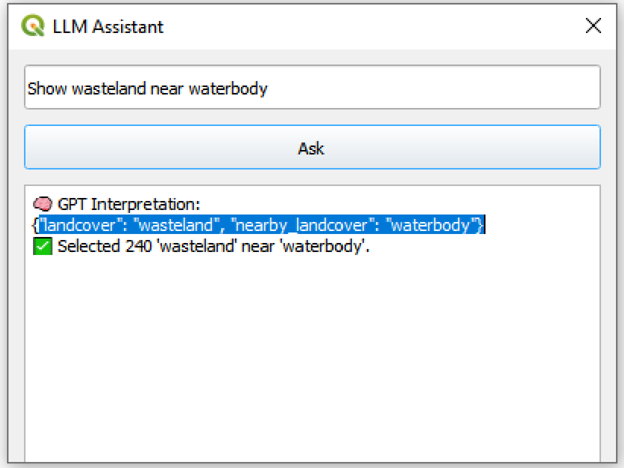


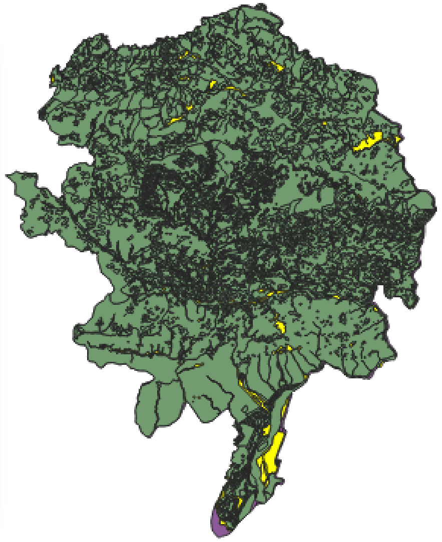


Natural Language GIS Queries using LLM –QGIS Plugin

Natural Language Query:
Show wasteland near waterbody

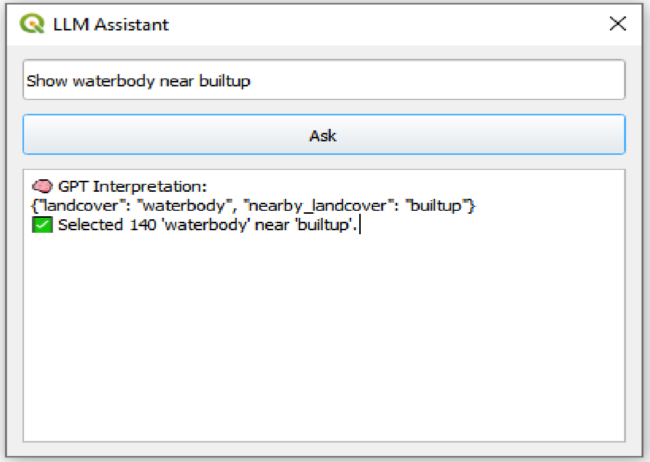
LLM Response:
jsonCopyEdit"landcover": "wasteland", "nearby_landcover": "waterbody"}

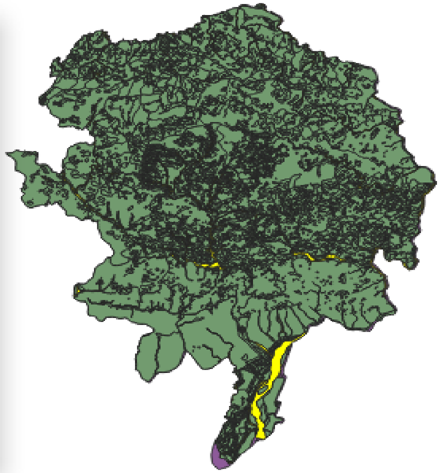




Natural Language Query:
Show wasteland near deep soil

LLM Response:
{"landcover": "wasteland", "soil_depth": "deep", "nearby_soil_depth": "deep"}
? Selected 180 'wasteland' on matching soil





OPEN PROBLEMS

Challenges & Limitations

Hallucination

Fluent, confident output that is factually or geometrically incorrect — risky for decision-critical spatial analysis.

Spatial Reasoning Gaps

Token-based models struggle with precise geometry, distances, and topology without external tools.

Domain Adaptation Cost

Fine-tuning and evaluation on geospatial schemas require curated data and domain expertise.

Compute & Latency

Larger models are expensive to host; real-time GIS workflows need efficient, often smaller, models.

Data Privacy & Governance

Sensitive government/geospatial data may require on-premise, air-gapped deployment.

Evaluation Difficulty

Measuring correctness of generated spatial queries/answers is harder than standard NLP benchmarks.

LOOKING AHEAD

Future Directions

- Geospatial foundation models: pretrained jointly on imagery, vector data, and text for unified reasoning.
- Multimodal LLMs that reason directly over maps, rasters, and point clouds alongside natural language.
- Agentic workflows: LLMs that plan and call GIS tools/APIs autonomously to complete multi-step spatial analyses.
- Tighter integration of retrieval, schema-grounding, and verification to reduce hallucination in high-stakes applications.

Where This Is Heading

The next generation of geospatial AI systems will likely combine LLM reasoning with dedicated spatial engines — using language as the interface, not the sole computation layer, for tasks that demand geometric precision.

Key skills to build now:

Prompt engineering · vector databases · spatial SQL · model evaluation · responsible AI practices

Key Takeaways

1. **Generative AI creates, not just classifies** — Built on deep learning and transformer architectures that learn rich data distributions.
2. **LLMs learn language through scale** — Pretraining, fine-tuning, and alignment turn raw text prediction into useful assistants.
3. **Prompting & RAG unlock new tasks** — In-context learning and retrieval let LLMs adapt without retraining from scratch.
4. **Geodata brings unique challenges** — Spatial reasoning, schema complexity, and precision demand careful grounding and evaluation.
5. **The field is moving toward multimodal, agentic systems** — combining language reasoning with dedicated geospatial tools and data.

Thank You

Questions & Discussion

Generative AI · LLMs · NLP · Geodata Processing